

NAJMEH NAYYER

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RESEARCH INTERESTS

Combining experimental and theoretical approaches to elucidate neural principles underlying behavior and cognition, particularly perception and decision-making, with the eventual goal of applying this knowledge to AI.

EDUCATION

B.Sc., Psychology, University of Isfahan, Isfahan, Iran Sep 2024 - Present
Current GPA: 18.41/20 (3.89/4) Major GPA: 18.17/20
Relevant Courses: *General Statistics, IT in Psychology, Research Methods, and Cognitive Psychology.*

SKILLS

Programming: Python and MATLAB
Packages: NumPy, Pandas, SciPy, Matplotlib, Seaborn, PsychoPy, MNE, Scikit-learn, EEGLAB
Other Software: MS Office (PowerPoint, Word, Excel), SPSS
Research Skills: Statistical Analysis, Experimental Design, Data Visualization, Signal Processing (Training)
Soft Skills: Project Management, Event Management, Mentoring, Leadership, Self-Study
Languages: Persian (Native), English (Advanced)

EXPERIENCE

RESEARCH EXPERIENCE

Research Intern May 2026 - Present

Supervisor: Prof. Mohammadreza A. Dahaghani & Dr. Farideh Shakerian, IPM

Independent Researcher

Sep 2025 - Present

Supervisor: Prof. Maryam Vaziri-Pashkam, University of Delaware

- Behavioral study on object recognition using object kinematograms.
- Formed the study hypothesis, organized the literature review and analyzed the initial pilot data.

Project Assistant

Apr 2025 - Aug 2025

Supervisor: Prof. Behrad Noudoost, University of Utah

[GitHub Link](#)

- Intraocular transfer of color adaptation under varying ocular conditions.

MENTORING EXPERIENCE

Mentoring Intern, Cognitive Neuroscience Competition for High School

Jan 2026 - Present

Iranian Neuroscience Society, Tehran, Tehran, Iran

Study Mentor for the National University Entrance Exam

Mar 2025 - Jul 2025

Saramaden e Rah e Danesh Institute (Notruphil), Hamedan, Iran

- Mentored high school students preparing for the exam and provided academic guidance and support.

PRESENTATIONS

Vahabi, F., Honarmand, F., **Nayyer, N.** (2025). *Chromatic Threshold Stability In The Yellow–Blue Axis*. Poster presented at the Basic and Clinical Neuroscience Congress, Tehran, Tehran, Iran ([Link](#))

SELECTED PROJECTS

Analysis of Natural Object Dataset EEG | Python

Ongoing

- Transformed data with Morlet wavelets, implemented machine learning algorithms, and compared activity in different classes of stimuli.

Python Bootcamp Projects ([GitHub Link](#))

Ongoing

- Developed 10+ interactive Python projects including games (Snake, Pong, Turtle Crossing) and utilities (Coffee Machine inventory system, Flashcard app), demonstrating OOP, GUI design, and data management.
- Built functional applications spanning game logic, inventory simulation, and study tools using Python fundamentals including loops, conditionals, classes, and event-driven programming.

Analysis of Psychology Class of 2028 Social Status Dataset | Python ([GitHub Link](#))

Ongoing

- Used NumPy, Pandas, SciPy, Matplotlib, and Plotly library to preprocess and analyze a survey from our sociology class.

Gender Inequality Attitudes Questionnaire | SPSS

Jul 2026

- Co-designed and validated a 30-item Likert-scale questionnaire for a Psychometrics course project.
- Analyzed data from 50+ participants, performing reliability analysis (Cronbach's $\alpha = 0.957$, Spearman-Brown = 0.978), indicating excellent internal consistency.
- Calculated descriptive statistics, standardized Z/T scores, and multiple cut-off score ($M + 2SD = 88.44$).

Memory-Guided Saccade Task | MATLAB [\(Project Link\)](#)

Jun 2026

- Analyzed multi-unit spikes (25 neurons) and LFP from macaque FEF during a memory-guided saccade task. (8 angles, 2 radii) Plotted raster and PTSH.
- Decoded target location from population firing rates using kNN and characterized LFP power spectra using multitaper method.
- Computed mutual information and AUC to quantify neural encoding of angle and radius across task phases.

Statistical Analysis Projects | SPSS

2025

- Two SPSS-based statistical analysis projects for the Practical Statistics and IT in Psychology course.
- Used descriptive statistics, correlation, linear regression, t-tests, ANCOVA, MANOVA, etc.

AWARDS & EXTRACURRICULAR

Brainies Book Club Founder and Leader [\(Link\)](#)

Jan 2026 - Present

- Presented “A Thousand Brains: A new Theory of Intelligence” by Jeff Hawkins for +10 students.

Dean’s List

Sem. 1 - Present

Vice President, Psychology students' association of University of Isfahan

Jun 2025 - Present

National Entrance Exam

2024

- Ranked 1135 (top 0.03%) among 348,917 participants nationwide which resulted in a full-ride scholarship.

SELECTED CERTIFICATES & OPTIONAL COURSES

Supervised Machine Learning: Regression and Classification, Andrew Ng, DeepLearning.AI (Coursera)	Nov 2025
Principles of Neuroscience, Behrad Noudoost, Isfahan University of Medical Sciences	May 2025
Data Structures, Charles Severance, University of Michigan (Coursera)	May 2025
AI for Everyone, Andrew Ng, DeepLearning.AI (Coursera)	Jan 2025
Understanding the Brain: Neurobiology of Everyday Life, Peggy Mason, University of Chicago (Coursera)	Dec 2024
Introduction to Data Analysis Using Excel, Sharad Borle, Rice University (Coursera)	Nov 2024